

Emory Hill

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EDUCATION

Georgia Institute of Technology

Atlanta, GA

B.S. Aerospace Engineering — BS/MS Honors Program (GPA: 3.85/4.00)

Dec. 2027

- Accelerated Master's Track in Aerospace Engineering
- Relevant Coursework: Aerodynamics, Jet & Vehicle Propulsion, Compressible Flow, Thermodynamics & Fluid Fundamentals, Control Systems & Analysis, Dynamics, Differential Equations

EXPERIENCE

Fabritek Aerospace Manufacturing (AS9100)

Winchester, VA

Aerospace Manufacturing Intern

May 2026 – Present

- Inspected **\$1M+** worth of aerospace components using metrology tools and CMM; caught a defect early in a **2,400-piece, \$50k** order, preventing **\$45,760** in scrap/rework across **2,200** remaining parts.
- Redesigned laser-engraving fixture to process **6×** more components per cycle, reducing operator time **67%** and increasing throughput **32%** across **\$800k+** worth of serialized components.
- Managed a customer-facing machining project (**±0.001 in** tolerances), producing CAD models, STEP/DXF files, engineering drawings, and travelers; delivered the full job **one week early**.

Georgia Tech Propulsive Landers

Atlanta, GA

Structures Engineer

Aug. 2024 – Present

- Iterated through **4** concepts and **5+** design versions for a **70 kg** rocket landing gear, reducing mass **24%** (3.46→2.63 kg) and cost **9%**, validated through FEA.
- Developed TPU honeycomb footpads, printing **20+** prototypes and increasing energy absorption **5.2×** (3.8→20 kJ) via Instron testing and MATLAB stress-strain analysis.
- Collaborated with propulsion and avionics teams to incorporate a **15°** TVC pitch requirement, determining minimum leg angle and structural safety factors.

PROJECTS

CFD Nozzle Flow Optimization | ANSYS Fluent, MATLAB

May 2026 – Present

- Conducting mesh-independence study across **3** meshes; optimized farfield biasing to reduce cell count **44%** (1.8M→1.0M) and cut iteration count **50%** (100k→50k), halving runtime.
- Simulating **12** nozzle geometries and a **6×3** test matrix (linear/bell contours across **3** ambient pressures), achieving exit Mach **3–4.5** using structured quadrilateral meshes.
- Automating Fluent post-processing in MATLAB to generate plots/tables; designing **6** additional optimized nozzles (data-based + theory-based) for thrust/efficiency comparison in final report.

MATLAB Quadcopter Dynamical Simulation | MATLAB

Sep. 2025 – Nov. 2025

- Achieved full 3D nonlinear quadcopter simulation fidelity by implementing a **12-state rigid-body dynamics model** using **ODE45**, measured through correct drag, inertia tensor, thrust distribution, and yaw-torque behavior
- Generated validated **3D trajectory** and orientation plots using custom MATLAB tools, measured by steady-state velocity matching and nonlinear behaviors consistent with real quadcopter dynamics.

AIAA Research Paper — Sloshing Nitrous Oxide | ANSYS Fluent

Oct. 2024 – Apr. 2025

- Simulated **9** baffle geometries (1–37 perforations), identifying optimal **7–13** hole range and improving damping ratio **59%** (0.536→0.852) for hybrid-rocket slosh mitigation.
- Performed steady-state Fluent analysis and validated simulation setup against theory; results will inform future GTPL hybrid-rocket baffle design.

TECHNICAL SKILLS

CFD & Simulation: ANSYS Fluent, ANSYS Meshing, ANSYS SpaceClaim, ANSYS Workbench, SolidWorks Simulation (FEA), MATLAB (ODE modeling, automation)

CAD & Design: SolidWorks (CSWP), Fusion360, Engineering Drawing (GD&T)

Programming & Data: MATLAB scripting, Python, Java, Excel

Manufacturing & Inspection: CMM inspection, laser engraving, machining process oversight, job travelers/documentation